

Shared Borrowers, Shared Stress: The Credit-Line Channel of Contagion

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Motivation

New York Times

NATIONAL EDITION

Rain and snow showers continue from Boston to Philadelphia as a storm moves off shore today. Showers and mountain snow persist in the West. Weather map, Page A24.

company

SATURDAY, MARCH 11, 2023

Prices in Canada may be higher

\$4.00



Labor Market Keeps Riding Its Hot Streak

*U.S. Adds 311,000 Jobs
as Some Sectors Slow*

By LYDIA DePILLIS

The labor market continued its energetic expansion in February, extending a hotter-than-expected streak that has created abundant job opportunities while frustrat-

U.S. TAKES OVER CRUCIAL LENDER IN TECH WORLD

BIGGEST BUST SINCE '08

Investors Batter Stocks of
Other Banks Catering
to Start-Up Clients

The bailout decision

- 13 March 2023: Silicon Valley Bank (SVB) collapses
NYT: “a financial contagion appeared to spread through parts of the financial sector [...] investors dumped stocks of [banks] [...] which cater to start-up clients”
- A failing bank forces a real-time decision: **bail out, or accept failure?**
- To decide, a regulator needs to know — *before markets react*:
 - which banks would be hit, and
 - by how much?

What we do

1. Introduce a **credit-line channel** of bank-to-bank contagion
 - two banks are linked if they lend to the same firms
 - “shared borrowers, shared stress”
2. Turn it into a **spillover matrix** and bank-level **systemicness**
3. **Compute** spillover matrix and systemicness
 - from supervisory loan-level data
 - (not using market prices)
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Why

- market-based measures rely on prices — pick up a crisis *too late*
- supervisory data quantify spillovers *ex ante*

Mechanism:

Shared borrowers, shared stress

Background: credit lines

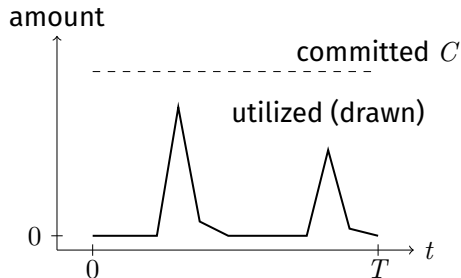
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 - $\sim 73\%$ of public firms
 - worth $\approx 16\%$ of book assets

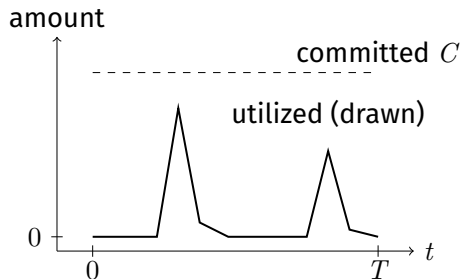
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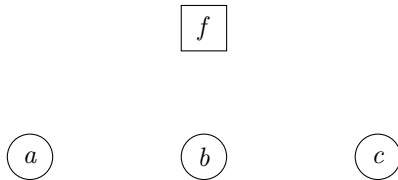


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- **large firms have credit lines with multiple banks** (Greenwald et al., 2025)

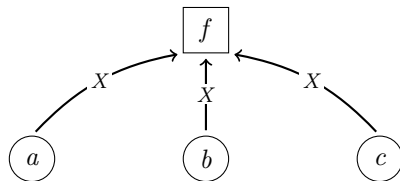


The credit-line channel of contagion



- firm f needs liquidity $3X$; it has credit lines at banks a , b , c

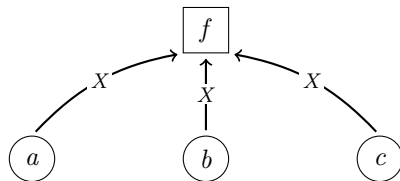
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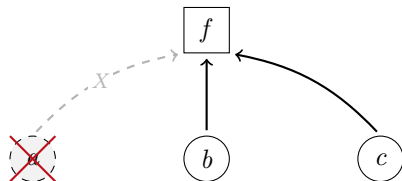
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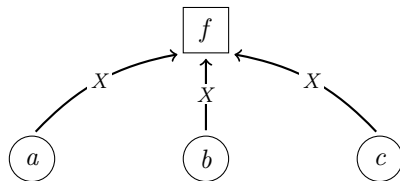
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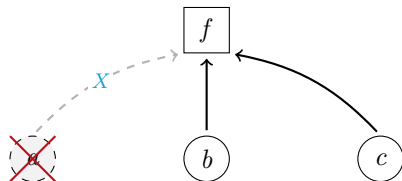
(b) bank a gone

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- if bank a fails

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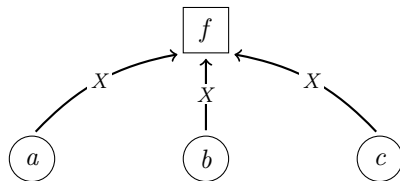
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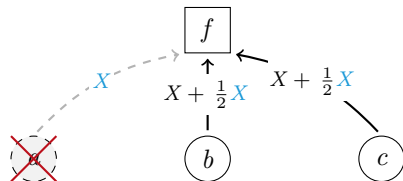
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- if bank a fails
 - firm f still needs that orphaned draw X

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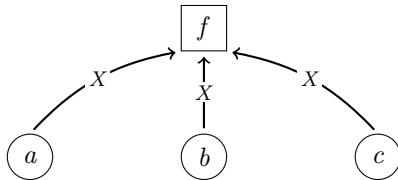
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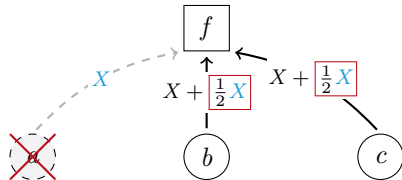
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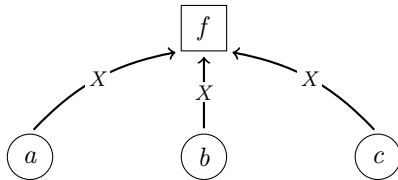
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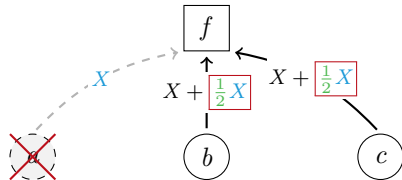
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 - **unexpected liquidity outflow**

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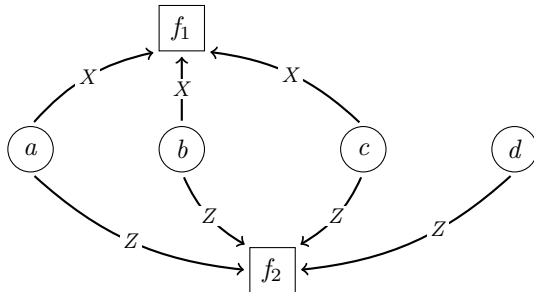
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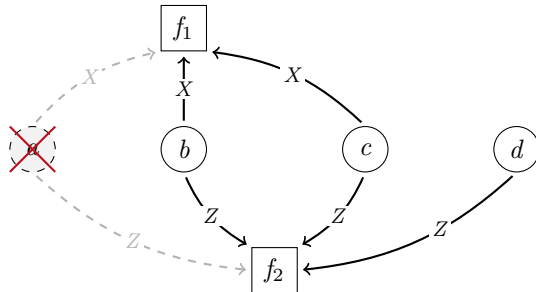
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 - **unexpected liquidity outflow** = **survivor share** $\times$ **orphaned draw**

Constructing bank-to-bank spillovers

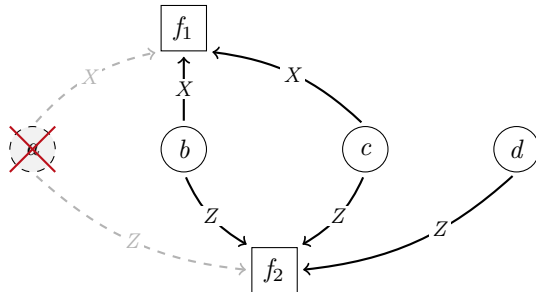


Constructing bank-to-bank spillovers



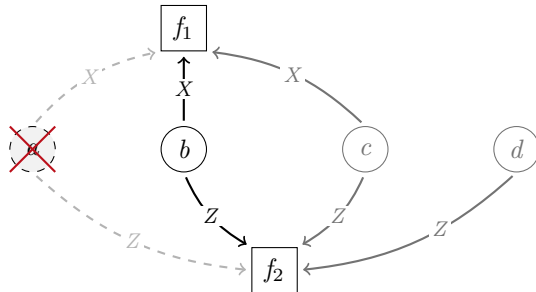
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Constructing bank-to-bank spillovers



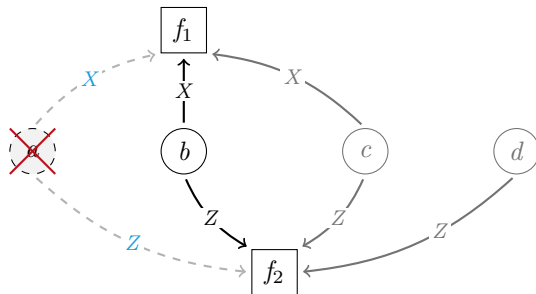
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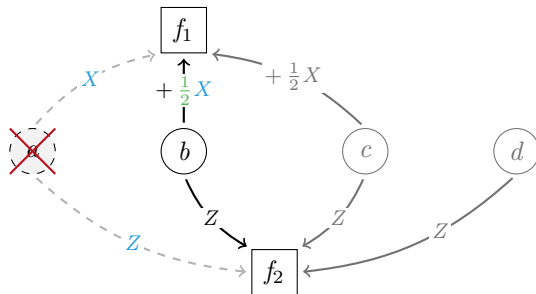
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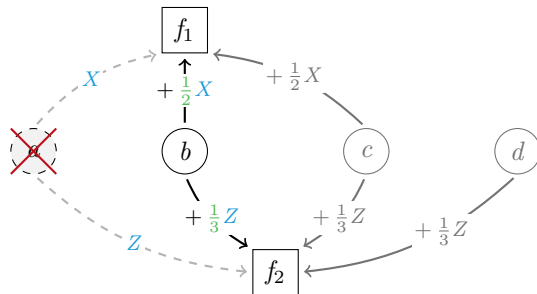
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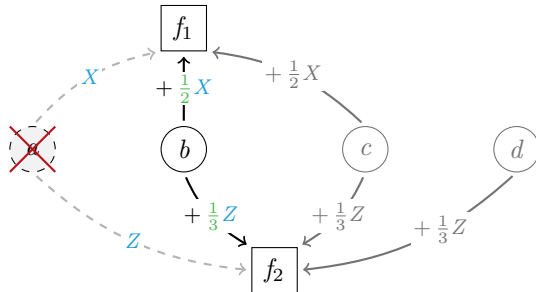
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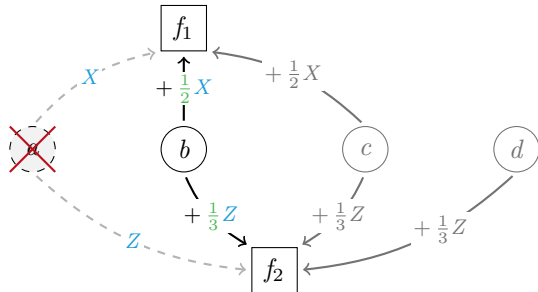
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- normalize by b 's buffer, $s_{ab} = \tilde{s}_{ab}/B_b$, and collect into S

The channel is real

- Lehman went down — its borrowers drew on their other credit lines (Ivashina and Scharfstein, 2010)

The channel is real and important

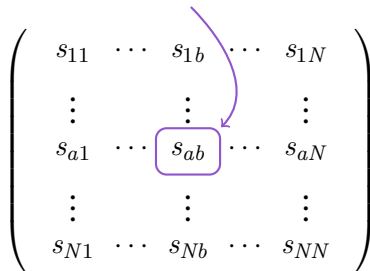
- Lehman went down — its borrowers drew on their other credit lines (Ivashina and Scharfstein, 2010)
- during the COVID crisis, banks with unexpectedly large drawdowns cut lending to everyone else (Greenwald et al., 2025)

Measuring systemic risk

The object: a spillover matrix S

- **spillover** s_{ab} : how a shock to a hits b — one entry of S

spillover $a \rightarrow b$



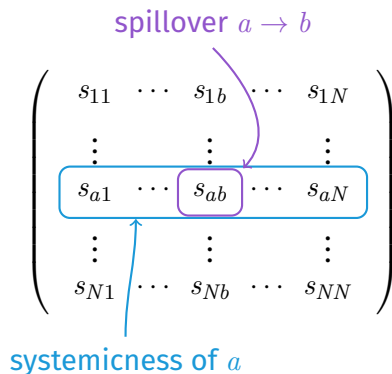
The diagram shows a matrix S with rows and columns indexed from 1 to N . The entry s_{ab} is highlighted with a purple box. A purple arrow points from the text "spillover $a \rightarrow b$ " to the box around s_{ab} .

$$\begin{pmatrix} s_{11} & \cdots & s_{1b} & \cdots & s_{1N} \\ \vdots & & \vdots & & \vdots \\ s_{a1} & \cdots & s_{ab} & \cdots & s_{aN} \\ \vdots & & \vdots & & \vdots \\ s_{N1} & \cdots & s_{Nb} & \cdots & s_{NN} \end{pmatrix}$$

Terminology after Greenwood et al. (2015).

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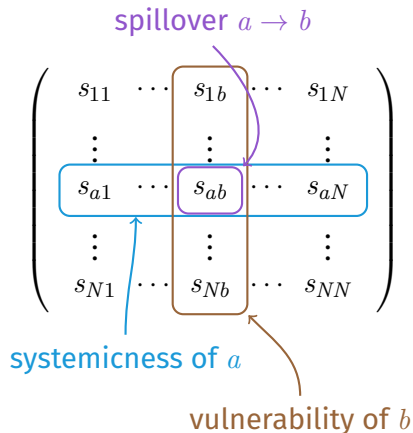


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- **vulnerability** of b : exposure to a system-wide shock — a *column* of S

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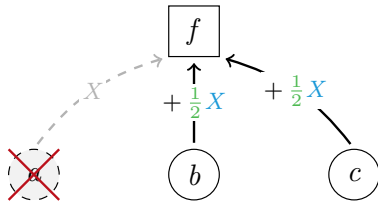
Who can measure it?

	Matrix S ?	Papers	Notes
<i>Reduced-form (equity co-movement)</i>			
equity returns (+ leverage)	aggregates	CoVaR, SRISK, MES	market data needed
equity-return VAR	full matrix	Diebold and Yilmaz (2014)	
<i>Structural (balance-sheet mechanism)</i>			
loans to banks	full matrix	Elsinger et al. (2006)* Denbee et al. (2021)	small
traded securities (fire sales)	aggregates*	Greenwood et al. (2015) Duarte and Eisenbach (2021)	
loans to firms	full matrix	this paper (credit lines)	

Computing the spillover matrix S

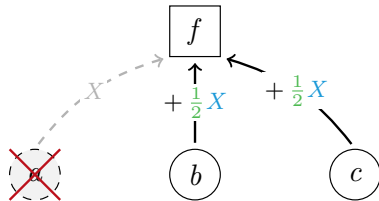
Three ingredients of the spillover s_{ab}

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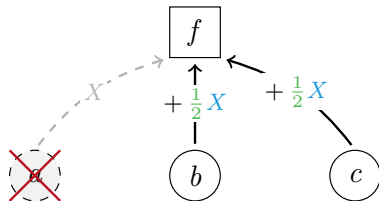
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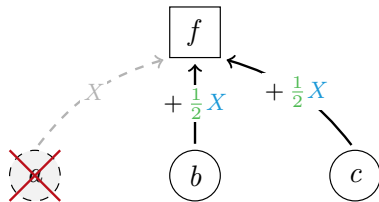
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 - baseline: in proportion to **headroom** (undrawn credit lines)

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 - baseline: in proportion to **headroom** (undrawn credit lines)
- **liquidity buffer** B_b : high quality liquid assets (HQLA)
 - main components are *observed*: cash, reserves, government bonds
 - regulation requires banks to hold enough HQLA to survive 30 days of stress

Data

Data

	AnaCredit	FR Y-14Q
coverage	euro area	United States
reporting banks	all banks	BHCs > \$100bn (≈ 20)
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- the **survivor share** needs *every* lender of a firm — twenty banks cannot deliver it
 - the **orphaned draw** is a **flow** — a quarter blurs what a month resolves
- build a bank $\times$ sector $\times$ month panel of **drawn** / **undrawn** credit lines

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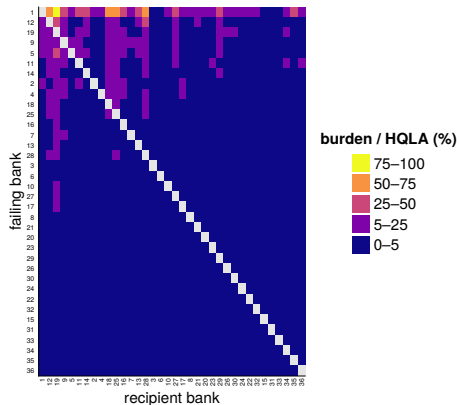
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- **This talk is about Austria** — the spillover matrix for the top 37 banks by HQLA

observations	banks	firms	sectors
instrument-months			NACE, 4 digits
7 458 894	446	114 772	651

Results: Austria

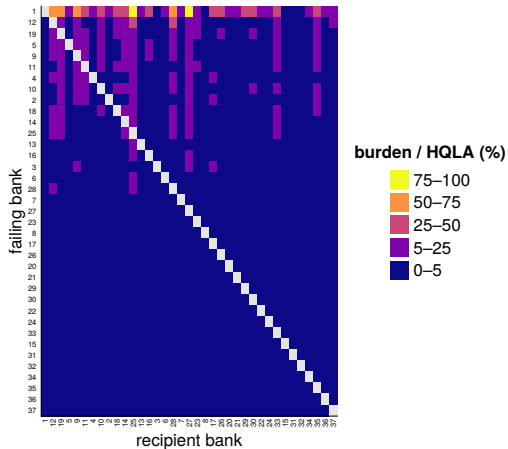
Computed spillover matrix: Austria, June 2025

“additional liquidity outflow at b if a defaulted, relative to b ’s HQLA”



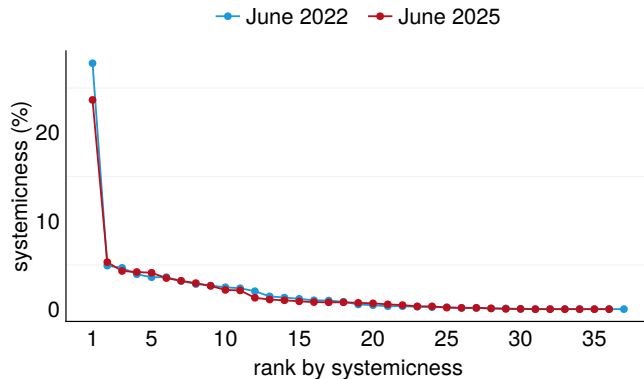
Notes: [preliminary](#) — this run uses drawn *balances* and splits in proportion to amounts *drawn*.
Top 37 banks by HQLA, ordered by systemicness (labels: HQLA rank).

The same picture three years earlier: June 2022



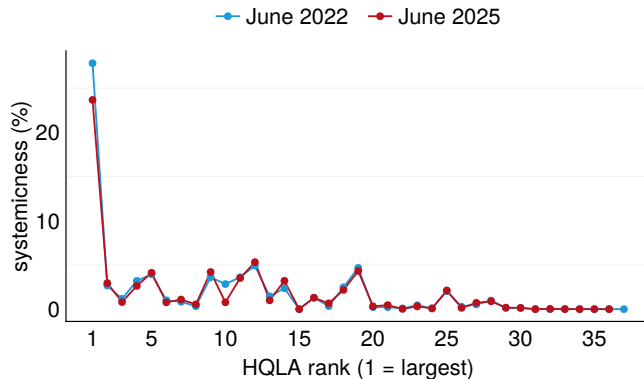
Notes: as on the previous slide, for June 2022.

Systemicness: average spillover of a bank's default



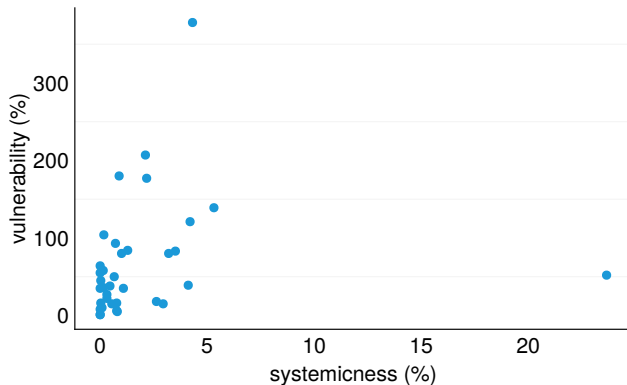
Notes: systemicness of a = row average of the spillover matrix: the mean outflow at the other banks if a defaulted, in % of their HQLA.

Systemicness by HQLA rank



Notes: the bank labels in the heatmaps are these ranks.

Systemicness and vulnerability

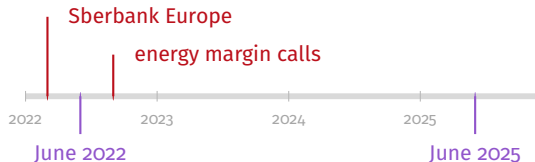


Notes: vulnerability of b = column sum: the outflow at b if all other banks defaulted, in % of b 's HQLA. June 2025.

Conclusion

Next steps

- Did the channel fire?
 - the 2022 episodes, monthly
- Horse race
 - credit line, interbank, fire sale



48 months in hand — two shown

Takeaways

- **Mechanism:** shared credit-line borrowers create a **structural** channel of bank-to-bank contagion
- **Measurement:** supervisory data alone build a spillover matrix and systemicness *ex ante*
- **Policy:** a **screen** for the support-vs-resolution decision — the latent tail that goes live the day no acquirer is found

References

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